Geometry

Lines

- Parallel lines do not meet.
- Intersecting lines meet at one point. The sum of any 2 adjacent angles formed is 180°.
- Perpendicular lines meet at one point, at 90°.

PRACTICE: Midpoint of a segment

(-2, 6) and (3, -2)

(12, 9) and (-2, -3)

Kinds of Angles

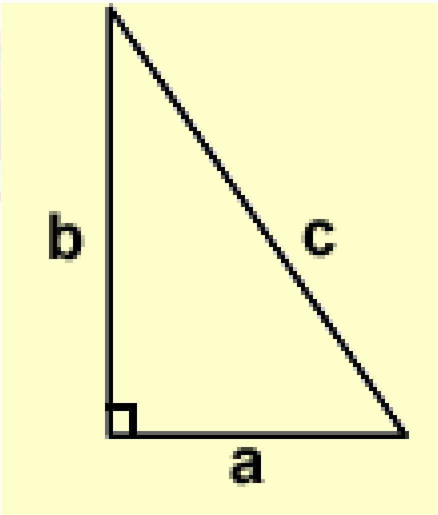
- Acute: less than 90°
- Right: 90°
- Obtuse: more than 90°, but less than 180°
- Straight: 180°
- Reflex: more than 180 degrees, less than 360°

Angle relationships

- Supplementary angles
  - Sum of measures is 180°
- Complementary angles
  - Sum of measures is 90°

Pythagorean theorem

- Applies to right triangles
- Legs: a and b
- Hypotenuse: c



$$c^2 = a^2 + b^2$$

PRACTICE

- John leaves school to go home. He walks 6 blocks north and then 8 blocks west. How far is he from the school?

Polygons

- Polygons are flat, closed figures consisting of line segments.
- Exterior angles of a polygon add up to 360°.
- The sum of the interior angles of a polygon, with the number of sides *n*: $180°(n-2)$
- The sum of the exterior angles of a simple, closed polygon is 360°.

Measurements

- Perimeter: sum of the length of all sides

$$P_{rectangle} = 2l + 2w \qquad P_{triangle} = a + b + c$$

$$Circumference = 2\pi r \qquad P_{square} = 4s$$

- Area: size covered by a surface

$$A_{square} = s^2 \qquad A_{rectangle} = lw$$

$$A_{circle} = \pi r^2 \qquad A_{triangle} = \frac{1}{2}bh$$

Measurements

- Volume: the space occupied or contained by a substance or shape

$$V_{cube} = s^3 \qquad V_{sphere} = \frac{4}{3}\pi r^3$$

$$V_{pyramid} = \frac{1}{3}(A \text{ of } b)h \qquad V_{cone} = \frac{1}{3}\pi r^2 h$$

$$V_{rectangular\ prism} = abc \text{ or } lwh$$

References and further reading

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